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May 14, 2026

Journal of Machine Learning Research

Dear Editors:

I am writing to submit my manuscript "Spectral Causality: Causal Direction Estimation via Magnetic Laplacians and Hodge Decomposition" to the *Journal of Machine Learning Research*.

The manuscript introduces *spectral causality*, a principled framework that estimates causal directions by exploiting the spectral structure of the magnetic Laplacian. I define a Directional Predictability Index (DPI) that resolves circularity in earlier utility-based formulations, connect the framework to Hodge decomposition to separate DAG-compatible and feedback components of causal flow, and establish scale-invariance and phase-transition results. The approach is validated on synthetic DAGs, the Sachs protein signaling network, and the UCI Heart Disease dataset, with benchmarks against DirectLiNGAM, PC, GES, and NOTEARS.

This paper has not been published previously in any journal or conference proceedings. A preprint may be made available on arXiv concurrent with this submission.

I suggest the following action editors and referees for this submission.

Action Editors:

- Bryon Aragam, University of Chicago (bragon@uchicago.edu) — causality, graphical models
- Elias Bareinboim, Columbia University (eb@cs.columbia.edu) — causal inference, generalizability
- Silvia Chiappa, DeepMind (silviac@google.com) — causal inference, variational inference
- Kenji Fukumizu, The Institute of Statistical Mathematics, Japan (fukumizu@ism.ac.jp) — kernel methods, dimension reduction
- Mladen Kolar, University of Southern California (mkolar@usc.edu) — graphical models, high-dimensional statistics

Reviewers:

- Joris Mooij, University of Amsterdam (j.m.mooij@uva.nl) — causal discovery, structural causal models
- Dominik Janzing, Amazon Tübingen (janzind@amazon.com) — causal inference, information-theoretic methods
- Biwei Huang, UC San Diego (bih007@ucsd.edu) — causal discovery, latent variable models
- Murat Kocaoglu, Purdue University (mkocaoglu@purdue.edu) — causal inference, graphical models
- Frederick Eberhardt, California Institute of Technology (fde@caltech.edu) — causal inference, Bayesian networks

This submission has the following keywords: causal discovery, magnetic Laplacian, Hodge decomposition, spectral graph theory, directed graphs.

Disclosure of funding and competing interests. No external funding was received for this work. The author declares no competing interests. The author has no conflict of interest with any of the suggested action editors or referees listed above.

Sincerely,

Tatsuki Onishi (Data Science and AI Innovation Research Promotion Center, Shiga University, Japan)